MUHAMMAD SHAHBAZ

Mianwali, Pakistan • bscs22f30@gmail.com • +92 302 4449316 • LinkedIn • GitHub

RESEARCH INTERESTS

> I am seeking MS research opportunities in cybersecurity, particularly machine learning for attack detection. I am interested in why intrusion detection models miss some attacks and raise false alarms on others, especially on noisy, heavily imbalanced traffic data. My directions include anomaly detection, concept drift, explainable intrusion detection, and cross-dataset evaluation.

EDUCATION

BS Computer Science — Namal University Mianwali, Pakistan

Oct 2022 – Jul 2026

CGPA: 3.19 / 4.00

Relevant Coursework: Machine Learning, Artificial Intelligence, Information & Cybersecurity, Computer Networks, Cloud Computing, Data Structures & Algorithms, Compiler Construction, Databases.

PUBLICATIONS

A Systematic Review of AI-Based Fake Job Detection (2017–2026)

Under Review **2026**

Submitted to *IEEE*. Systematic review of AI-based approaches to fake job posting detection published between 2017 and 2026.

Manuscript PDF available on GitHub repository.

HONORS & ACHIEVEMENTS

National Skill Competency Test (NSCT)

Apr 2026

Higher Education Commission (HEC), Pakistan — **98.5th percentile (Top 1.5%)** among 33,038 candidates nationwide. Evaluated across core domain competencies including programming, databases, AI/ML, cybersecurity, and algorithmic problem solving.

FEATURED RESEARCH PROJECT

ML-Based Network Intrusion Detection

2026

- Developed a comprehensive machine-learning pipeline for binary and multiclass attack classification on the benchmark CIC-IDS2017 dataset, rigorously evaluating Logistic Regression, Random Forest, HistGradientBoosting, and XGBoost under multiple class-imbalance mitigation strategies.
- Applied leakage-aware data preprocessing, feature-importance analysis, and error analysis. Evaluated performance with metrics including precision, recall, macro F1, and ROC-AUC scores.
- In preliminary binary experiments, XGBoost reached an impressive **macro F1 score of 0.9976** with a low false-negative rate of 0.38% on a 200,000-row stratified subset.

- Investigated class imbalance, rare attack vectors, benchmark dataset artifacts, and generalization bounds to establish groundwork for continuous anomaly detection and concept drift monitoring.
- **Tools & Libraries:** Python, scikit-learn, XGBoost, pandas, NumPy, Matplotlib, Seaborn, Hypothesis.

WORK EXPERIENCE

Software Engineering Intern — Speridian Technologies

Jul – Aug 2025

- Worked at a multinational technology consulting firm operating across nine countries, participating actively in professional software engineering workflows, Agile sprints, and team code reviews.
- Engineered enterprise software solutions using C# and .NET for scalable backend microservices and RESTful APIs, integrated with Angular and TypeScript for dynamic web applications.
- Designed optimized database schemas and executed complex query optimizations in Microsoft SQL Server to ensure seamless performance.

TECHNICAL & ENGINEERING PROJECTS

Hybrid Encryption Secure Chat Application

Python / Security

Architected a multi-client secure chat application utilizing hybrid cryptographic protocols (RSA-2048 key exchange combined with AES-256-GCM message encryption). Included features for user-managed keys and real-time step-by-step visual inspection of encryption/decryption routines.

AI-Powered Disaster Management System

Final Year Project

Developed an end-to-end full-stack emergency response platform (Angular, Node.js, MongoDB) featuring 4 role-based management dashboards (Victims, NGOs, Admin, Volunteers). Integrated a machine learning predictive model forecasting resource requirements 3 days in advance alongside real-time geospatial mapping using Leaflet.js.

Soybean Yield Prediction Model

Machine Learning

Constructed a statistical regression model to predict agricultural yields based on environmental and crop traits, applying rigorous feature selection, exploratory data analysis (EDA), and statistical modeling.

TECHNICAL SKILLS & CERTIFICATIONS

Languages:	Python, C#, JavaScript, TypeScript, SQL, HTML/CSS
ML & Cybersecurity:	Machine Learning, Deep Learning, Network Intrusion Detection, Model Evaluation (F1, ROC-AUC), Applied Cryptography (RSA, AES-GCM), scikit-learn, XGBoost, IoT Security Fundamentals
Frameworks & Tools:	Angular, RxJS, .NET, Node.js, Express.js, RESTful APIs, SQL Server, MySQL, MongoDB, Leaflet.js, Bootstrap, Git, Postman, pandas, NumPy, Matplotlib, Seaborn
Certifications:	Introduction to Cybersecurity (Cisco), Introduction to IoT (Cisco), Intro to Deep Learning (Kaggle)
Language Fluency:	English (Professional Working Proficiency), Urdu (Native)